CSS Selectors

A COMPLETE GUIDE

Full Stack Development Course
for Beginners

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CSS selectors are patterns used to select and style HTML elements.

They select one or more elements based on tags, classes, IDs, attributes, states, or relationships in the document structure.

BASIC SELECTORS

These selectors select elements directly by their tag name, class, or ID.

Universal Selector (*)

Selects all elements on the page.

```
/* Apply zero margin and padding to every element */
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}
```

Element Selector

Selects all elements of a specific type.

```
/* Style all <p> elements */
p {
  line-height: 1.6;
  color: #333;
}
```

Swipe

.class

Selects all elements with a specific class name.

```
/* Style any element with class="btn" */
.btn {
  background: #007bff;
  color: white;
  padding: 0.5rem 1rem;
  border-radius: 4px;
}
```

#id

Selects a single element with a matching ID.

```
/* Style the element with id="main-header" */
#main-header {
  font-size: 2rem;
  text-align: center;
}
```

COMBINATORS

Combinator selectors select elements by tracing their relationships, such as parent-child or siblings.

This lets you style elements based on their position in the DOM hierarchy.

A B

Selects all B elements that are descendants of A.

```
/* Style all <p> elements inside <article> */
article p {
  font-size: 1rem;
}
```

A > B

Selects B elements that are direct children of A.

```
/* Style <li> elements that are direct children of a <ul> */
ul > li {
  list-style-type: none;
}
```

A + B

Selects the first B element immediately following A.

```
/* Style the first <p> after an <h2> */
h2 + p {
  margin-top: 0;
}
```

A ~ B

Selects all B siblings that come after A.

```
/* Style all <p> elements that are siblings after an <h3> */
h3 ~ p {
  color: gray;
}
```

ATTRIBUTE SELECTORS

These selectors match elements based on the value of HTML attributes, providing a flexible way to target elements like inputs, links, or images without needing classes or IDs.

[attr]

Selects elements that have the specified attribute.

```
/* Style all inputs with the "disabled" attribute */
input[disabled] {
  background: #eee;
  cursor: not-allowed;
}
```

[attr="value"]

Selects elements with an exact attribute value match.

```
/* Style all checkboxes */
input[type="checkbox"] {
  margin-right: 8px;
}
```

[attr~="value"]

Selects elements where the attribute contains the value as a separate word.

```
/*
Style elements that have "highlight"
as one of their class names
*/

div[class~="highlight"] {
  background: yellow;
}
```

[attr^="value"]

Selects elements whose attribute value starts with the given value.

```
/* Style links that begin with "https" */
a[href^="https"] {
  color: green;
}
```

```
/* Style images ending with .png */
img[src$=".png"] {
  border: 2px solid #ccc;
}
```

[attr$="value"]

Selects elements whose attribute value ends with the given value.

[attr*="value"]

Selects elements whose attribute value contains the value anywhere.

```
/* Style links whose title includes "docs" */
a[title*="docs"] {
  font-weight: bold;
}
```

PSEUDO-CLASS

Pseudo-class selectors target elements in a specific state or position, such as when a user hovers over a link, or when an element is the first child in a container.

:hover

Targets an element when a user hovers over it.

```
/* Add underline on hover for links */
a:hover {
  color: #e60023;
  text-decoration: underline;
}
```

:focus

Targets an element when it gains focus.

```
/* Highlight input field when it's focused */
input:focus {
  border: 2px solid #007bff;
}
```

```
/* Add a click effect to buttons */
button:active {
  transform: scale(0.98);
}
```

:active

Targets an element when it's being clicked.

:first-child

Targets the first child of a parent element.

```
/* Make the first list item bold */
ul li:first-child {
  font-weight: bold;
}
```

:last-child

Targets the last child of a parent element.

```
/* Style the last item in a list */
ul li:last-child {
  color: red;
}
```

```
/* Apply zebra striping to table rows */
tr:nth-child(even) {
  background-color: #f2f2f2;
}
```

:nth-child(n)

Targets the nth child of a parent.

:nth-of-type(n)

Targets the nth element of its type within a parent.

```
/* Style the second paragraph in a container */
p:nth-of-type(2) {
  font-style: italic;
}
```

:not(selector)

Selects all elements that do not match the selector.

```
/* Hide all divs except the ones with class="visible" */
div:not(.visible) {
  display: none;
}
```

Swipe

:checked

Targets checked inputs like checkboxes or radios.

```
/* Bold label when checkbox is checked */
input:checked + label {
  font-weight: bold;
}
```

```
/* Dim the appearance of disabled buttons */
button:disabled {
  background: #ccc;
}
```

:disabled

Targets disabled form inputs.

:empty

Selects elements with no child nodes.

```
/* Show an empty state block */
div:empty {
  height: 20px;
  background: #eee;
}
```

PSEUDO-ELEMENT

Pseudo-element selectors style specific parts of an element's content, such as the first letter, first line, or content inserted before or after the element.

::before

Inserts content before the element's actual content.

```
/* Add a hash symbol before headings */
h1::before {
  content: "# ";
  color: gray;
}
```

```
/* Add sparkle after paragraph */
p::after {
  content: " ✨";
  color: gold;
}
```

::after

Inserts content after the element's actual content.

::first-letter

Targets the first letter of the element's text.

```
/* Enlarge the first letter of paragraphs */
p::first-letter {
  font-size: 2rem;
  color: #007bff;
}
```

::first-line

Targets the first line of the element's text.

```
/* Bold the first line of each paragraph */
p::first-line {
  font-weight: bold;
}
```

::placeholder

Styles the placeholder text inside input or textarea.

```
/* Style placeholder text */
input::placeholder {
  color: #999;
}
```

::selection

Targets the first line of the element's text.

```
/* Highlight selected text */
::selection {
  background: yellow;
  color: black;
}
```